THOMPSON RIVERS UNIVERSITY

Department of Engineering

Faculty of Science

SENG 4620 - Practical Cloud Computing

Design Project: Building a Scalable and Global Cloud Infrastructure for a Local Business

This is a group project with a maximum of 3 students per team.

This semester, you'll go on a journey to build, secure, and scale the cloud infrastructure for a local business called *The Riverside Cafe*. Starting with no web presence, you'll guide The Riverside Cafe through a complete digital transformation by solving real-world business challenges at each step. The project is divided into eight distinct phases, each with its own deliverable.

Learning Objectives

- To architect a cloud infrastructure that meets evolving business requirements for scalability, security, cost-effectiveness, and reliability.
- To design and implement a defense-in-depth network security strategy by building a custom Virtual Private Cloud.
- To define and deploy an entire cloud infrastructure as code (IaC) using a version-controlled, automated CI/CD pipeline.
- To design a fault-tolerant architecture that uses load balancing and auto-scaling to handle dynamic traffic loads across multiple Availability Zones.
- To design and implement an event-driven, serverless workflow to solve a business problem in a cost-effective and operationally efficient manner.
- To synthesize and communicate complex technical architectures and design decisions through professional documentation, diagrams, and presentations.

The Project - Phase by Phase

Phase 1: The Launch and Global Presence (Due: End of Week 3)

- **Business Challenge:** The Riverside Cafe has no website and is invisible online.
- **Your Task:** Launch a simple, cost-effective, and reliable **static website** to put The Riverside Cafe on the map. You'll also establish data protection and disaster recovery plans from day one. (Note: The website files are provided to you by a web developer and are available on Moodle as “Phase 1 - Static Website”.)

Phase 2: Dynamic Website with Online Ordering (Due: End of Week 4)

- **Business Challenge:** Customers want to place orders online. The static site can't handle this.
- **Your Task:** Evolve the architecture to host a **dynamic web application** on a virtual machine within a secure network that can process customer orders. (Note: The website files will be provided to you by a web developer.)

Phase 3: Building a Resilient Database (Due: End of Week 5)

- **Business Challenge:** The on-server database is a performance bottleneck and a management nightmare (backups, patching).
- **Your Task:** Decouple the database by migrating it from the web server to a **managed, relational database service** while improving security and resilience. (Note: The original database information and credentials will be provided by a database engineer.)

Phase 4: Fortifying the Network - Advanced Security (Due: End of Week 6)

- **Business Challenge:** The application server is still too exposed to the internet, and secure administrative access is needed.
- **Your Task:** Design and build a new, highly **secure network** from scratch, isolating servers in private subnets and implementing a bastion host for secure management access.

Phase 5: Architecting for Scale (Due: End of Week 8)

- **Business Challenge:** The Riverside Cafe is about to be featured on a TV show, anticipating a massive traffic spike that will crash the single server.
- **Your Task:** Re-architect the application tier for high availability and elasticity using a **load balancer and an auto-scaling group** across multiple Availability Zones.

Phase 6: Full Automation - Infrastructure as Code & CI/CD (Due: End of Week 9)

- **Business Challenge:** Manual deployments are slow, error-prone, and inconsistent, hindering expansion.
- **Your Task:** Define your entire infrastructure as code (**IaC**) and build an automated **CI/CD pipeline** to deploy changes reliably and quickly.

Phase 7: Optimizing with Serverless - Smart & Efficient (Due: End of Week 10)

- **Business Challenge:** A new daily sales report task is slowing down the main website and running a dedicated server for it is too expensive.
- **Your Task:** Solve the problem with a modern, event-driven approach. Build a cost-effective **serverless workflow** that runs the task on-demand without impacting performance.

Phase 8: Project Documentation and Presentation (Due: End of Week 12)

- **Business Challenge:** The project is a success, and now the work must be formally documented and presented for handover and knowledge sharing within the organization.
- **Your Task:** Create a comprehensive **final report**, a presentation, and a video demonstration that encapsulates the entire project journey and technical architecture.

Project Constraints

Throughout all phases of this project, your designs and implementations must consider the following cross-cutting constraints:

- **Cost-Effectiveness:** The Riverside Cafe is a small business with a limited budget. You must justify your selection of services and instance sizes based on a balance of performance and cost. Avoid over-provisioning.
- **Reliability:** The Riverside Cafe's website is a critical business tool. Your architecture must be resilient to failures, with clear plans for data protection and disaster recovery.
- **Security:** Your top priority is to protect The Riverside Cafe's data and infrastructure. You must adhere to the principle of least privilege, secure sensitive information, and build defense-in-depth.
- **Scalability:** While starting small, your designs should always consider future growth. The architecture must be able to scale efficiently to meet changing demands.

Teamwork

This project will be completed in groups of 2-3 students. You must inform the instructor of your team members via email by the **end of the first week**. For students who have not formed a team, the instructor will randomly create groups and post them on Moodle on the first day of the second week.

The most important part of a successful team is finding a time when everyone meets, as a group, each week, for at least two hours. Each student should expect to spend about four hours outside of class per week on the project. Compare your schedules to find a time block of at least two hours when you all can meet. Commit to meeting at this time each week and working together. If something comes up and someone can't make it, be sure to schedule another time for that week.

You will deliver your team charter during the first phase of the project. A team charter is a document of your own design. It should be “artfully” designed, expressing some interests and passions of your team. Things to discuss: What are your goals for the class? Who are your team members? What is your team mascot? How will the team celebrate triumphs? How will the team make important decisions? How will the team resolve conflicts and discuss problems? What does “leadership” mean to your team? Who is the person that hits “submit” on your reports and milestones? What are the skills of the team members? When will the team meet as a group each week? (Please be precise.) What will be the procedure for missing or being late to this meeting? How much advance notice must be given and using what method? A team charter template has been uploaded alongside the project files.

Assessment

This project is divided into eight phases. Each phase is detailed in a separate document. Phases 1 through 7 are each worth 100 marks. Phase 8 is worth 200 marks. The marking rubric for each phase is provided at the end of its respective document. Your total score will be out of 900 marks, which will then be scaled to a final project grade out of 100.